IIT PALAKKAD

Design of Optical Systems for Hybrid Photoluminescence Imaging, Spectrometer and Dynamic Moiré Sensor

Project II : End-Term Report

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Abstract

This work presents three optical sensing systems developed for various applications including **Hybrid Photoluminescence Imaging**, **Spectrometer** and **Dynamic Moire Sensor based Displacement Measurement**.

The first project focuses on a **Hybrid Photoluminescence Inspection System** developed to evaluate solar cell quality by detecting cracks and defects through optical excitation. Photoluminescence imaging is a non-contact optical technique used to analyze recombination activity and identify defects in photovoltaic devices.

The second project focuses on the development of a **Spectrometer** for optical spectrum measurement across different applications. One key application is Spectral Compensation for a Solar Panel Level IV tester, where spectral mismatch between the illumination source and the standard solar spectrum at which it is tested can introduce measurement errors.

The third project focuses on the development of high-resolution and long-range displacement sensor based on **Dynamic Moiré Pattern** using Linear CMOS Array Sensor and electronically tunable grating pattern.

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Part I

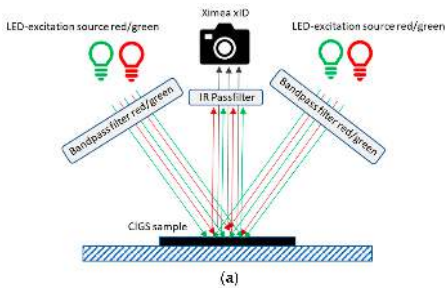
Hybrid Photoluminescence Inspection System

Chapter 1

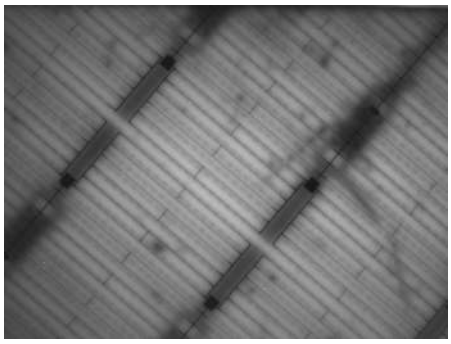
Introduction

1.1 Background

Photoluminescence (PL) imaging has emerged as a non-contact technique for the spatial characterization of semiconductor materials and solar cells. When a silicon wafer or solar cell is illuminated with photons of sufficient energy, excess carriers are generated and radiative recombination, emitting luminescence that can be captured using an imaging system [9, 11]. The captured image can be represents the micro cracks and defects in the material. The intensity of the emitted photoluminescence is directly related to important material parameters such as minority carrier lifetime, diffusion length, and recombination activity[10, 13, 12, 14].



(a) PL Imaging System [15]



(b) PL Image of Solar Cells

Figure 1.1: Photoluminescence Imaging Setup and Image

1.2 Project Objectives

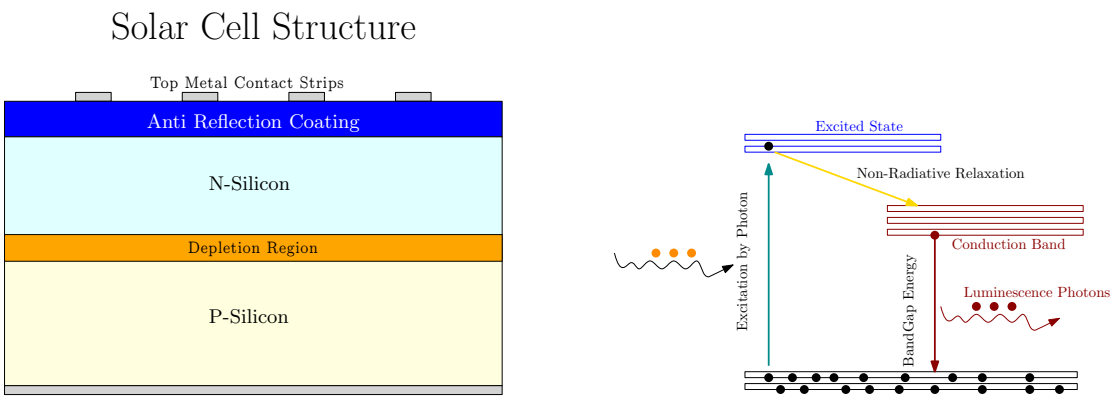
The objective of this project is to develop a Hybrid Photoluminescence System for Solar Cells with illumination from both the top side and bottom side to detect defects in the surface layers and the bulk regions of the solar cell.

Chapter 2

Literature Review

2.1 Photoluminescence Imaging of Solar Cells

A standard crystalline silicon (c Si) solar cell consists of a p-n junction formed by a n type emitter layer and a p type base layer, effectively creating a large area diode structure. To maximize light absorption and overall efficiency, the top surface is coated with an Anti Reflective Coating (ARC). Electrical current is collected via a front metal contact grid consisting of fine finger strips connected to larger bus bars and back metal contact 3.9a



(a) Structure of Solar Cell (b) Luminescence

Figure 2.1: (a) Solar cell structure and (b) Luminescence working principle.

When the silicon base is excited, electrons in the valence band absorb the incident photons and transition to an excited state in the conduction band, creating electron-hole pairs. These excess carriers subsequently return to a lower energy state through recombination. In high-quality silicon, a portion of this process occurs via **radiative recombination**, where the energy is released as near-infrared photons peaking at approximately 1150 nm. The intensity of this emitted light serves as a spatial map of the carrier lifetime, as regions with high defect concentrations gives non-radiative recombination which helps us to identify defects easily Fig. 2.1b.

The excitation can be done electrically by passing current through the solar cells which is called **Electroluminescence**, or by photons incident to the solar cells which is **Photoluminescence**.

Chapter 3

Challenges, Experiments & Analysis

3.1 Photoluminescence Imaging System

Photoluminescence Imaging setup is shown in the Fig.3.1 where an LED light source is illuminated to Solar Cell and it absorbs the light and emits photoluminescence spanning from approximately 950nm-1300nm with a peak at 1125nm corresponding to bandgap of 1.1eV.

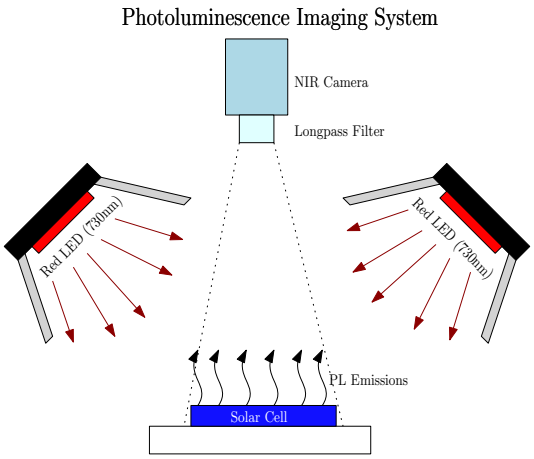


Figure 3.1: Photoluminescence Setup

3.2 Cameras for PL Imaging

Cameras used to measure Photoluminescence are SWIR(Short Wave InfraRed) Cameras which is having high sensitivity in wavelength range of 900-1700nm. But these cameras are very costly as it uses InGaAs Image sensors instead of CCD Detectors.
NIR Cameras are much cheaper comparison to SWIR Cameras, but their Quantum Efficiency falls off to zero near 1100 nm as shown in Fig. 3.2. So the effective area of sensing is reduced to 850nm-1100nm (shaded region)

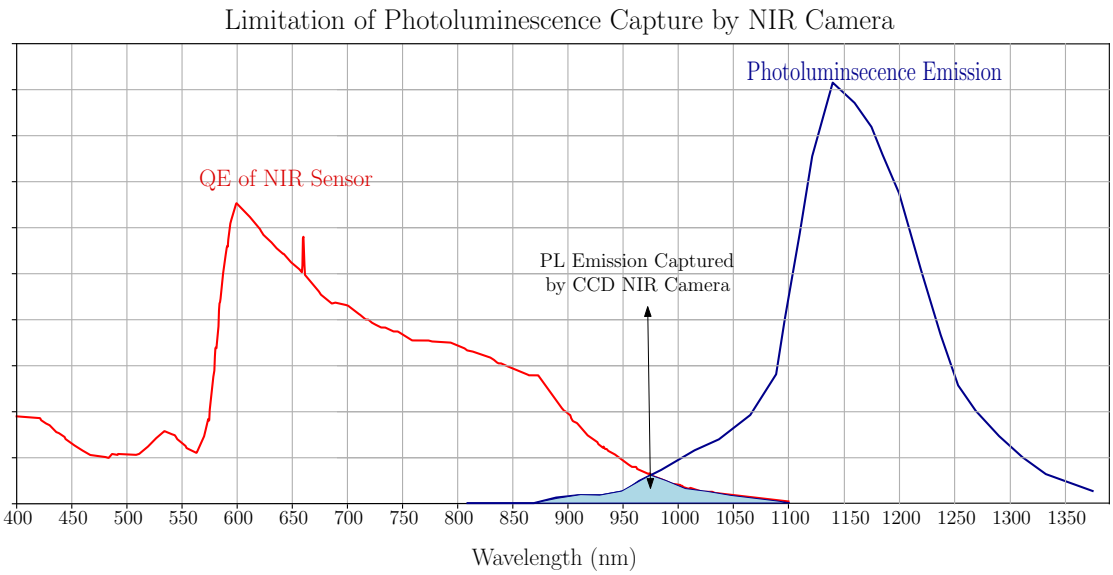


Figure 3.2: QE of Camera v/s PL Emission Spectrum

3.3 730nm LED Array for Excitation

For Exciting the electrons, a 730nm LED Array was used which was having a spectrum as shown in Fig. 3.3. The peak intensity of LED Spectrum is above $300\mu W/cm^2/nm$, The clipping can be seen as the spectrometer was saturated at $300\mu W/cm^2/nm$

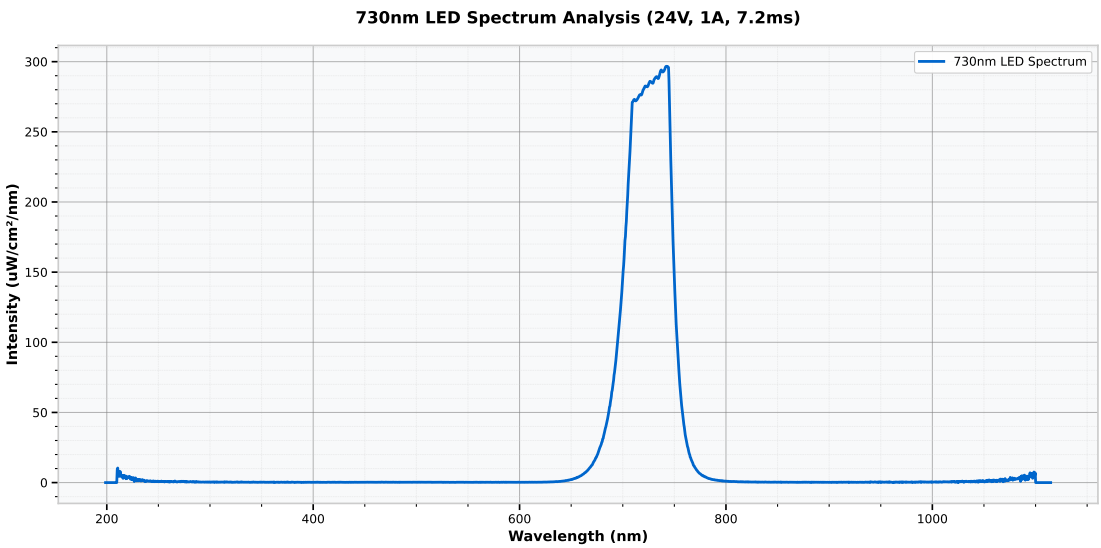


Figure 3.3: Spectrum of LED

For comparison, we have taken the Photoluminescence emission spectrum and Electroluminescence spectrum using Ocean Optics Spectrometer Fig .3.4 We can see that the intensity of PL and EL emissions are below $2\mu W/cm^2/nm$ in the range of 1100nm, which is the end point of QE of camera. If the incident LED Spectrum of $300\mu W/cm^2/nm$ is not filtered out, then PL images will have reflected light with dominant intensity

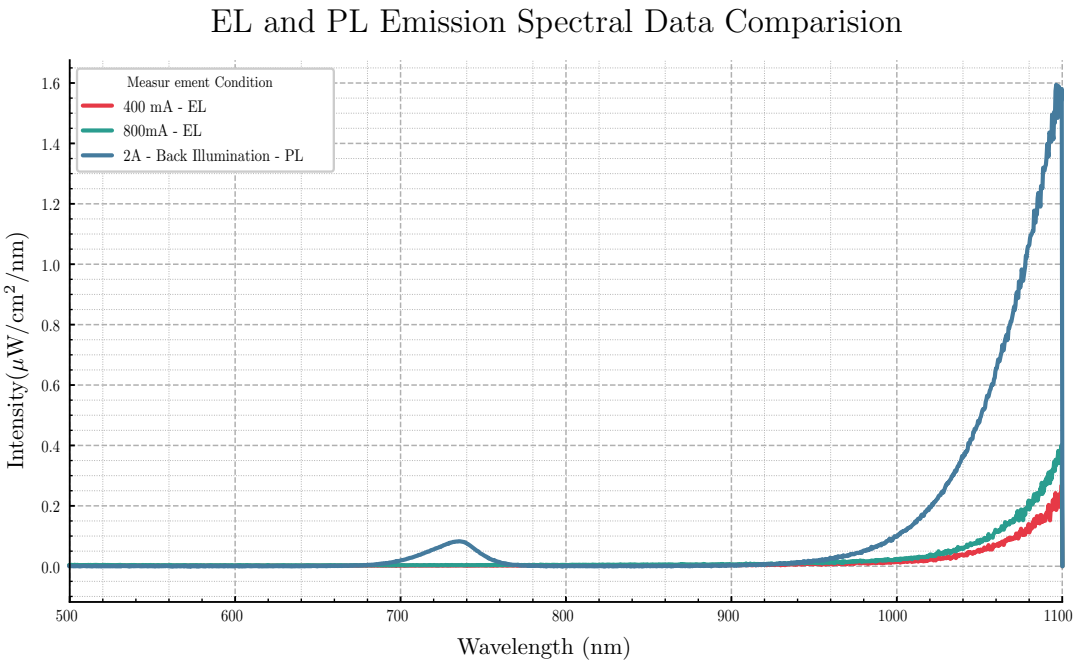


Figure 3.4: EL and PL Spectrum

3.4 Optical Filter

In order filter out the incident 730nm LED light from camera, we used a long pass filter front of Image Sensor with cutoff frequency of 875nm and Optical Density of 2.0. Filter Response is shown in Fig 3.5

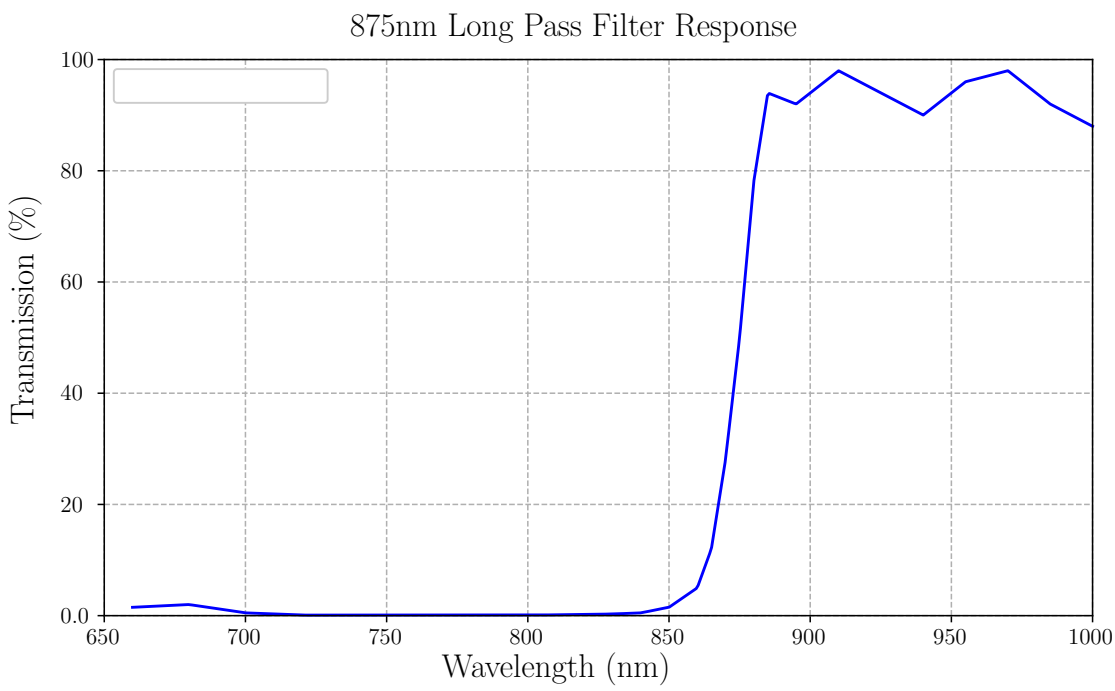


Figure 3.5: Long Pass Filter cutoff at 875nm

With the above optical arrangement, we captured the image of Solar Cell with NIR Camera, expecting incident light is cutoff completely. But the results we got 3.6, where reflected light was getting dominated instead of PL emissions. From

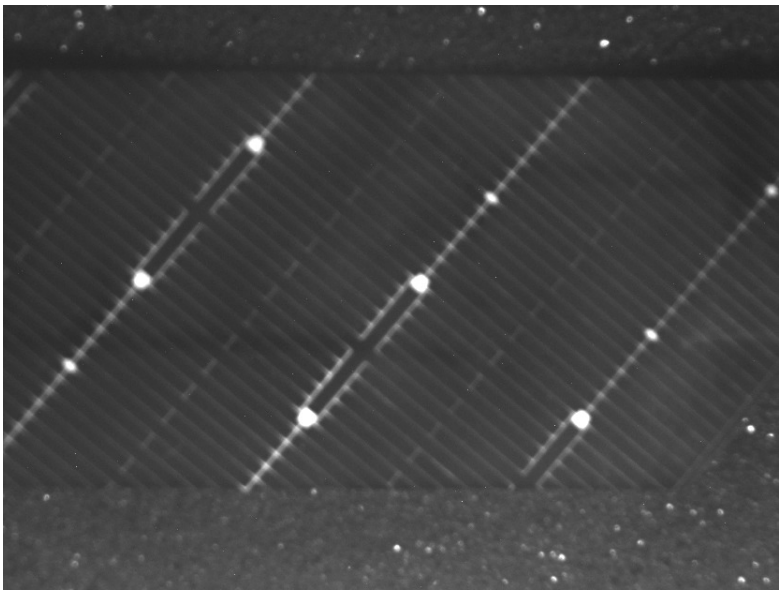


Figure 3.6: Image of Solar Cell, Reflection Dominated PL emission

the datasheet of the Optical Filter, the average OD is given as 2.0

$$T = 10^{-OD} = 10^{-2.0} = 0.01 = 1\% \tag{3.1}$$

1 % transmission of intensity greater than $300\mu W/cm^2/nm$ is greater than $3\mu W/cm^2/nm$, which confirms that LED Reflection dominated PL emissions due to filter leakage.

3.5 Spectral Analysis of Filter Leakage

With the incident light and OD 2.0, 875nm Long pass filter we have taken the spectrum from Solar Cell on the direction of Camera Fig. 3.8. This spectrum confirms that LED Reflection was dominating with around $50\mu W/cm^2/nm$ which suppressed the PL Emission of $2\mu W/cm^2/nm$

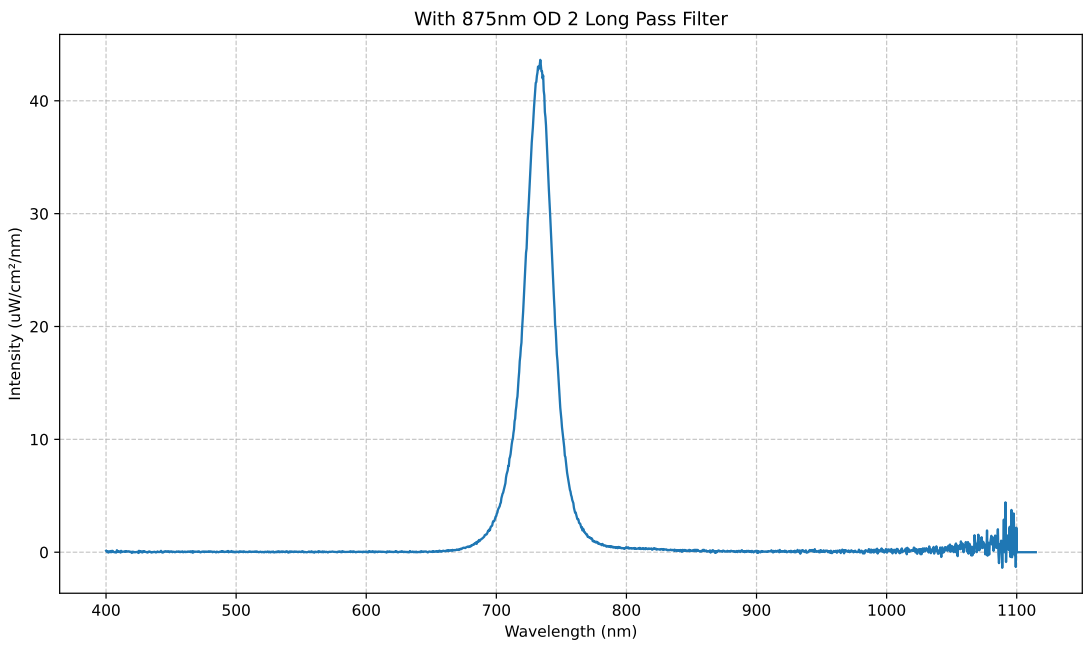


Figure 3.7: With OD2.0 Filter

Next, a spectrum was taken utilizing the silicon solar cell itself as a primary optical filter. Crystalline silicon is a semiconductor with an indirect bandgap of approximately 1.12 eV at room temperature. Consequently, it exhibits strong absorption for all wavelengths shorter than its bandgap (approximately λ¡1100 nm). This is done by illuminating from the backside of solar cells. Now the intensity reduces to $3 - 4\mu W/cm^2/nm$ and PL emissions were seen fairly near 1050nm and above.

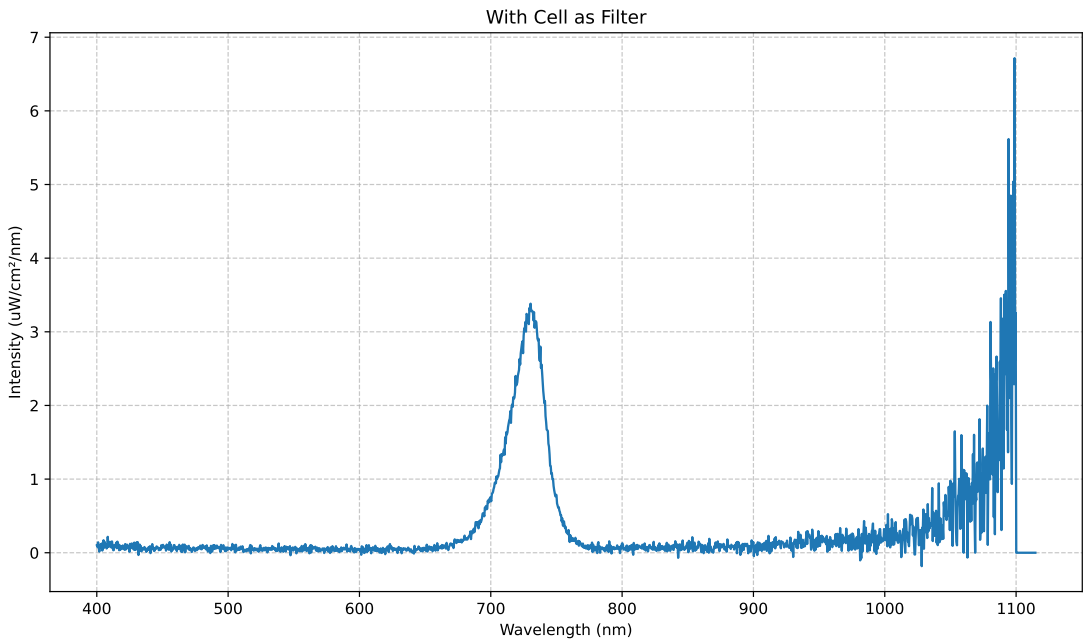
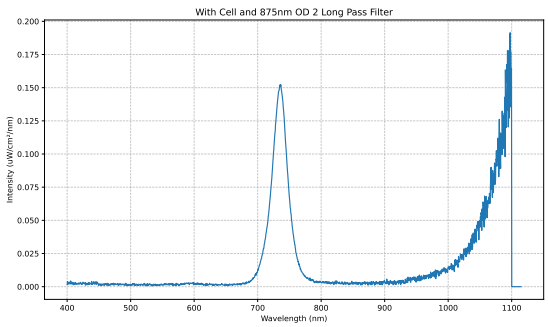
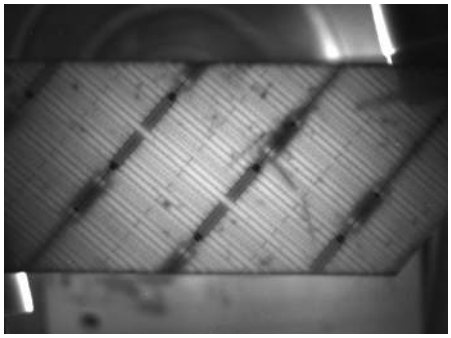


Figure 3.8: With Solar Cell as a filter

Combining Silicon Filtering by back illumination and OD 2.0 Filter, we were able to cutoff the incident light properly and obtain the PL emissions Fig. 3.9. Photoluminescence image as obtained in Fig 3.9b, with backside illumination have cracks/defects and metallisation in lower intensity (black color), whereas solar cells without defects having higher intensities (white color)



(a) Spectrum of Combined Filter



(b) PL with Back Illumination

Figure 3.9: Back Illuminated Spectrum and PL Image

3.6 Hybrid Photoluminescence Imaging

A hybrid photoluminescence system featuring dual-side illumination is proposed to overcome the limitations of standard single-sided PL imaging. By providing excitation from both the top and bottom surfaces, the system can distinguish between localized surface defects and global bulk properties.

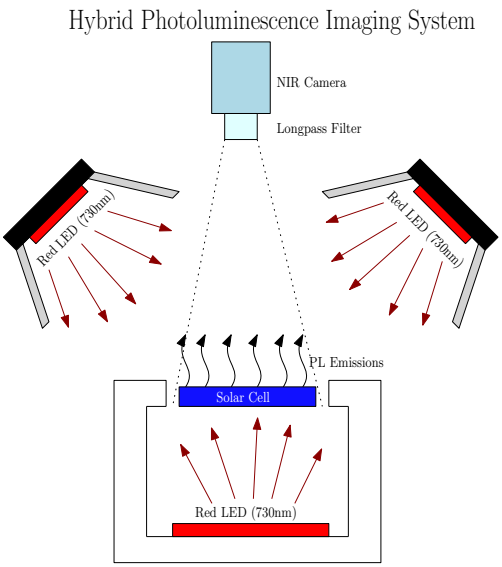


Figure 3.10: Hybrid PL setup

Chapter 4

Results

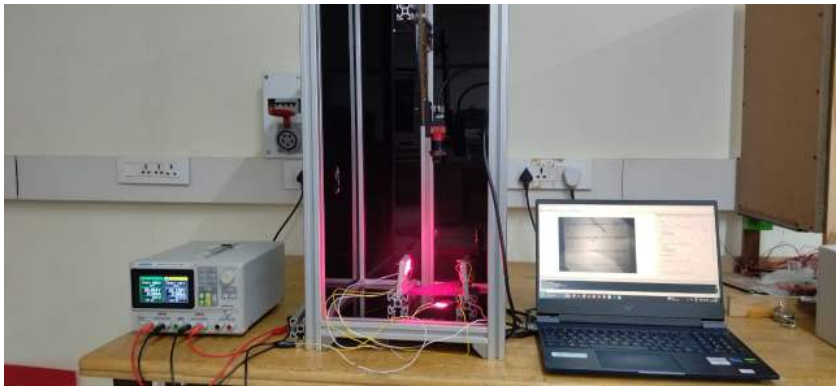
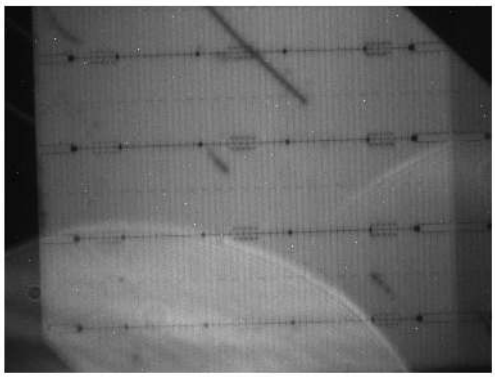
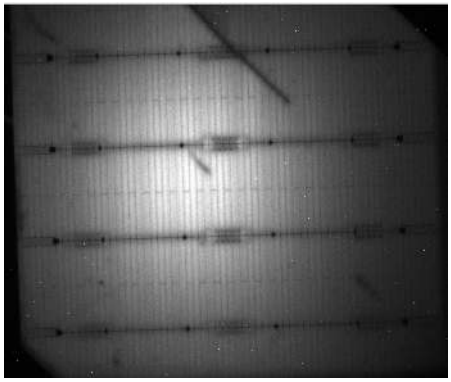


Figure 4.1: Test System for Hybrid PL

Figure 4.1 shows the experimental setup used for hybrid photoluminescence (PL) measurements. A 1000 nm cutoff OD 5.0 long-pass filter is used to suppress the excitation wavelength, and its tail response extending down to around 950 nm enables partial capture of the PL emission.



(a) Top Illumination



(b) Bottom Illumination

Figure 4.2: Photoluminescence Results under Different Illumination Conditions

Chapter 5

Conclusion and Future Perspective

Initial experiments demonstrated that an OD 2.0 filter alone cannot suppress high-intensity 730 nm LED reflections, which leak at $45\mu W/cm^2/nm$ and drown out the $2\mu W/cm^2/nm$ PL signal.

By implementing backside illumination, the silicon substrate acts as a natural longpass filter, absorbing excitation light while remaining transparent to the 1150 nm PL emission.

To address this, backside illumination was implemented, where the silicon substrate effectively acts as a natural long-pass filter—absorbing the excitation wavelength while allowing the $\tilde{1}150$ nm PL emission to pass through. This approach significantly improves signal isolation and measurement quality. The R&D phase of the hybrid PL system has been successfully completed, and future work will focus on integrating both photoluminescence (PL) and electroluminescence (EL) techniques into a unified characterization platform

Part II

Design of Spectrometer

Chapter 1

Introduction

1.1 Background

Accurate spectral measurement is essential in systems involving spectral analysis. One of the common application is evaluating solar module performance under varying irradiance conditions. The incident solar spectrum varies with atmospheric composition, time of day, and geographic location. Solar module datasheets typically provide I–V curves under standard test conditions (STC), corresponding to the AM1.5 spectral distribution and 1000 W/m^2 irradiance. However on-site panel level IV Curve tracer will have deviation in their irradiance measurement, causing difference in IV Curves of Solar Modules.

This project aims to develop two mini spectrometer modules: one based on the Hamamatsu modules of VIS and NIR spectrum, another one using the industry - standard Czerny - Turner optical configuration.

1.2 Project Objectives

- Mini-Spectrometer Device with modules from Hamamatsu
- Czerny-Turner Spectrometer with wavelength range from 300nm-1000nm

Chapter 2

Theory of Spectrometers

2.1 Diffraction Grating

A diffraction grating is an optical element consisting of a large number of equally spaced grooves. When light is incident on the grating, each groove acts as a secondary source of light waves. The interference between these waves causes different wavelengths of light to diffract at different angles.

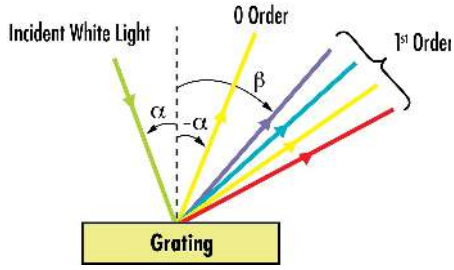


Figure 2.1: Diffraction from a grating

The condition for constructive interference is given by the grating equation

$$d(\sin\alpha + \sin\beta) = m\lambda \tag{2.1}$$

where d is the spacing between adjacent grating grooves, α is the angle of incidence, β is the diffraction angle of order m, and λ is the wavelength of the incident light.

As each wavelength satisfies this condition at a different angle, the grating separates polychromatic light into its spectral components. This principle forms the basis of wavelength dispersion in spectrometers.

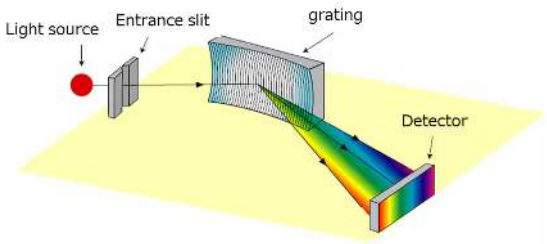


Figure 2.2: Working of Spectrometer

A basic spectrometer configuration consists of a **slit** to limit the light entering into the device, a diffractive element (**eg: diffracting grating**) to diffract the incident lights into wavelengths spatially and a **sensor** to detect the intensity of each light and converting into voltage waveform Fig 2.2

There are several optical configurations that can be used to obtain spectral measurements as shown in Fig 2.2. Among these, the crossed Czerny–Turner configuration was selected for this project. Along with that, spectrometer modules from Hamamatsu is also used for developing a Mini-Spectrometer module.

2.2 Crossed-Czerny Turner Spectrometer

Crossed Czerny - Turner Spectrometer is a widely used optical configuration for compact spectrometer design.

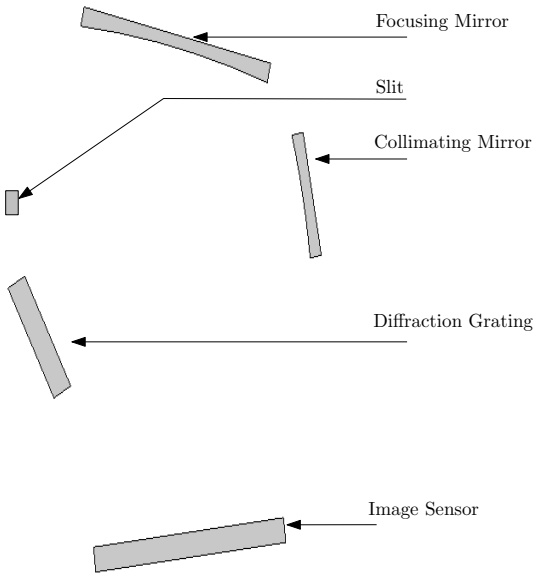


Figure 2.3: Crossed Czerny-Turner Configuration

The main components of the crossed Czerny–Turner spectrometer are:

- Entrance Slit (10 - 200 μm width)
- Collimating Mirror (Collimates input light)
- Diffraction Grating (With Specified groove density)
- Focusing Mirror (Focuses to Image Sensor)
- Image Sensor

The entrance slit with specified opening width will allow controlled amount of light to enter the system. The entrance slit is placed at the focus of Collimating mirror, so collimating mirror will make the incident conical light rays into a parallel beam of light and will shine it onto diffraction grating. The groove density (n lines/mm)

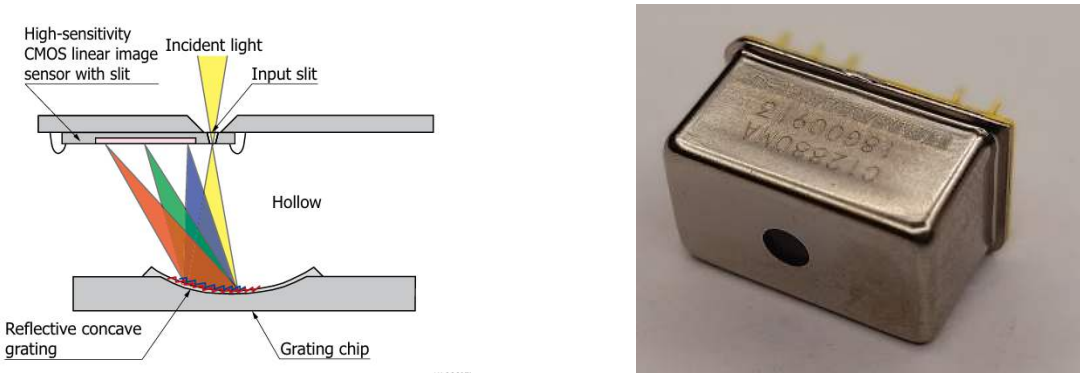
of the diffraction grating will follow the equation and will diffract each wavelength in different angles

$$\sin\alpha + \sin\beta = nm\lambda \tag{2.2}$$

where n - (grooves/mm) of diffraction grating, m - order of diffraction, α - angle of incidence, β - angle of reflection. The focusing mirror gets the parallel diffracted light rays and it focuses it to image sensor and we can read out the intensity of light at each wavelength from the image sensor

2.3 C12880 Spectrometer

Along with the development of the Czerny–Turner configuration spectrometer, a compact spectrometer module based on the Hamamatsu C12880MA is also used in this work.



(a) Internal Structure of C12880

(b) C12880 Spectrometer Module

Figure 2.4: C12880 spectrometer and its internal structure

The Hamamatsu C12880MA is an ultra-compact mini spectrometer module designed for portable spectral measurement systems.

It integrates an entrance slit, concave diffraction grating which diffracts and focuses light in to a high-sensitivity CMOS linear image sensor within a small package. The device measures light in the wavelength range of approximately 340 nm to 850 nm

Chapter 3

Spectrometer Design

3.1 Optical Simulation of Crossed Czerny-Turner Configuration

Optical design and simulation of Crossed Czerny Turner Configuration is done with COMSOL Multiphysics Software. The design uses 2 Mirrors, each of 25mm diameter and 50mm focal length and diffraction grating of groove density 600 lines/mm. Photo-detector of 3648 pixels, of length 30cm is used to detect the spectrum in this simulation. We will be incorporating the distances between each objects according to this simulation while designing the Spectrometer

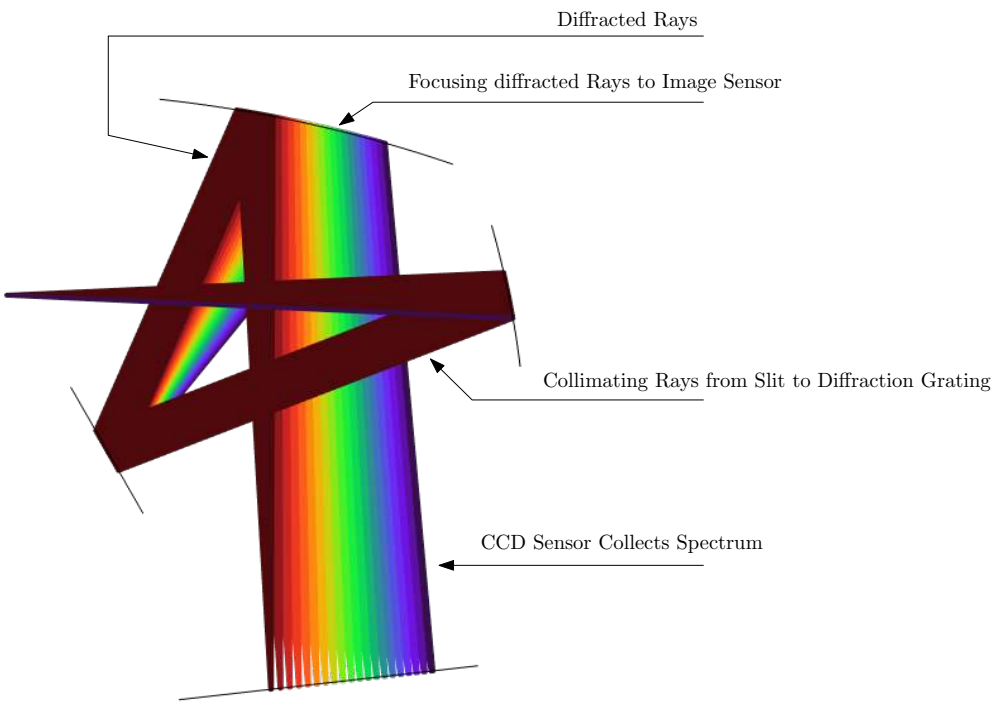
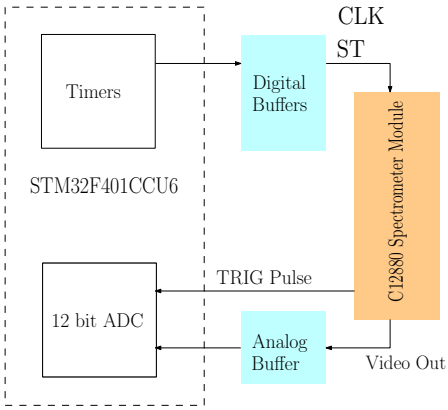


Figure 3.1: Simulation in COMSOL Multiphysics

3.2 C12280 Spectrometer

A compact spectrometer system based on the Hamamatsu C12880 mini-spectrometer module was implemented as part of this work. To interface with the sensor and meet its timing requirements, a dedicated driver PCB was designed using the STM32F401CCU6 microcontroller. The microcontroller generates the required clock and start pulses for sensor operation and controls the data acquisition process. The analog video output from the spectrometer is sampled using 12 bit ADC. Complete System is shown in Fig 4.1b

(a) C12880 Image Sensor



(b) C12880 Driver Circuit

Figure 3.2: C12880 Spectrometer and its driver circuit

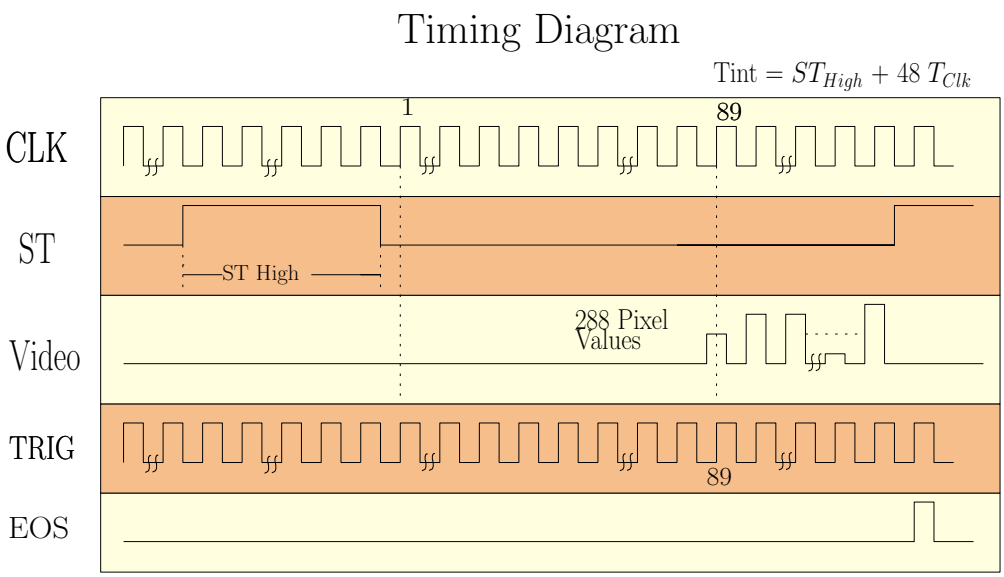


Figure 3.3: Timing Diagram for Spectrometer

The timing requirements for the C12880 spectrometer is shown in Fig. 3.6. A 1 MHz clock and a start pulse are applied to control the sensor operation and set the integration time. The integration time is determined by the high duration of the start pulse. The sensor provides three output signals: Video, TRIG, and EOS (End of Scan). The analog pixel values from all 288 pixels are sequentially shifted out through the video output and must be sampled at the positive edge of the TRIG signal. After all 288 samples are read, the EOS signal generates a high pulse for one clock cycle, indicating the completion of the scan. The shifted and sample 288 pixel values can be calibrated and converted into spectrum corresponding to 340nm - 850nm

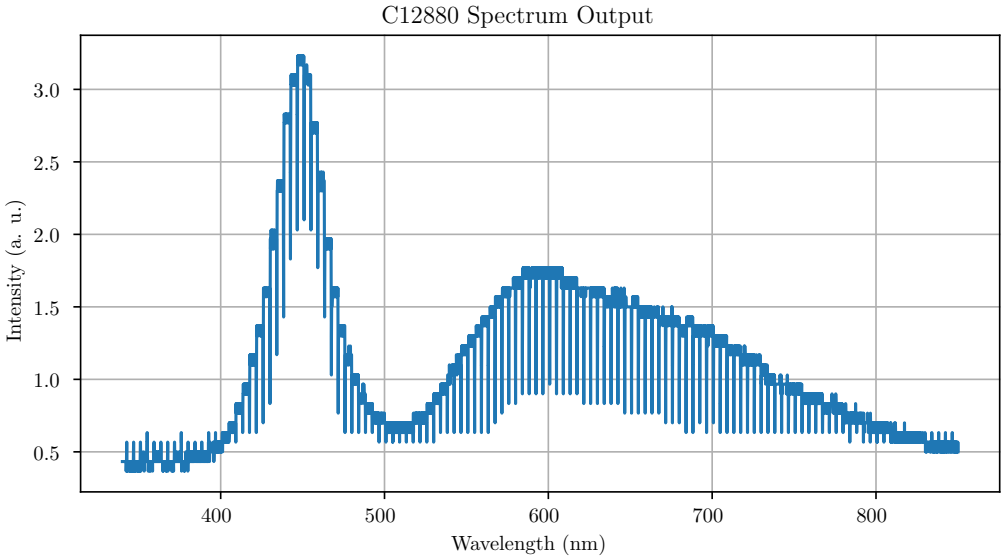


Figure 3.4: White Light Spectrum

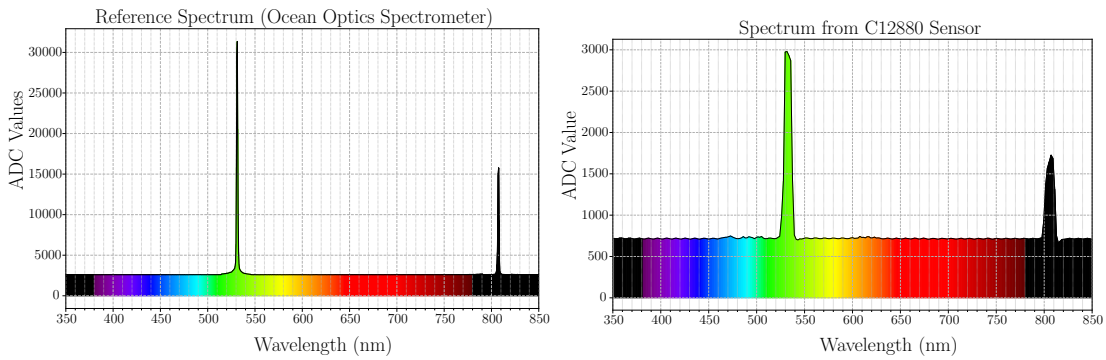
In Fig 3.4, the shifted out pixels values viewed in oscilloscope and converted time axis to wavelength

Chapter 4

Experiments and Results

4.1 C12280 Spectrometer

To evaluate the performance of the developed spectrometer, the measured spectral output was compared with the reference spectrum obtained from the **Maya2000Pro spectrometer from Ocean Optics**, which is a calibrated laboratory-grade spectrometer. The comparison was performed using a 531 nm green light source. The measured spectrum shows a dominant peak around 531 nm along with an additional peak beyond 800 nm in the near-infrared region. This secondary peak occurs due to the inherent NIR emission component present in the light source. By comparing the spectra obtained from both instruments, the wavelength alignment and spectral response of the developed spectrometer can be validated. From the results, we can observe that the spectral resolution of the C12880 spectrometer is lower compared to the reference spectrum, since the number of wavelength samples in the 340–850 nm range is limited to 288, whereas the Ocean Optics spectrometer provides more than 1000 samples over the same range. However, the C12880 spectrometer is still capable of providing an approximate spectral distribution and correctly identifying the major wavelength peaks.



(a) Reference Spectrum

(b) Spectrum from C12880

Figure 4.1: Reference spectrometer and the C12880 spectrometer

Chapter 5

Conclusion and Future Perspective

Two architectures were evaluated: a portable Hamamatsu C12880 module and a custom Czerny-Turner design. The Hamamatsu module, driven by an STM32F401CCU6, successfully identified spectral peaks (e.g., 531 nm) despite its lower 288-pixel resolution. COMSOL simulations of the Czerny-Turner configuration confirmed that a 600 lines/mm grating can provide high-resolution dispersion for a 300–1000 nm range. Future efforts will focus on manufacturing the optomechanical housing for the Czerny-Turner prototype.

- Development and prototyping of Czerny Turner Spectrometer
- Calibration and Standardizing C12880 Spectrometer

Part III

Dynamic Moiré Displacement Sensor

Chapter 1

Introduction

1.1 Background

In displacement measurement, non-contact optical methods such as interferometry and Moiré-based techniques helps in detecting small deformations. The *Sampling Moiré Method [3]* enables precise displacement mapping by analyzing phase differences in periodic patterns, providing submicron resolution without physical contact.

However, conventional implementations are limited by static gratings and restricted measurement ranges. By introducing dynamic, electronically tunable gratings generated using LCD displays, high resolution and long range can be achieved.

1.2 Project Objective

Development of Optical Sensor module for Dynamic Moiré displacement Sensor using CMOS Linear Image sensor and LCD display which can be used for high-resolution, long-range displacement measurement

Chapter 2

Literature Review and Theory

2.1 Moire Pattern Based Displacement Sensing

The Moire effect is a phenomenon that occurs when two or more patterns with similar but slightly different frequencies or phases overlap. As a result of the constructive and destructive interference of the patterns, a resultant pattern emerges Fig. 2.1 The resultant pattern is related to the type of mismatch between the patterns (E.g. frequency, angle, displacement)

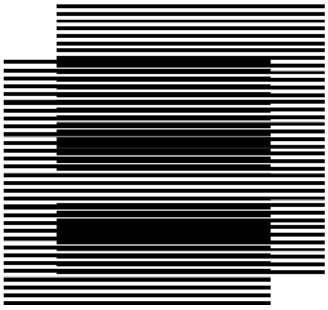


Figure 2.1: Moire Pattern [2]

2.2 Sampling Moire Method

The Sampling Moiré Method is a non-contact optical technique used to measure small deformations or displacements of an object with high accuracy. In this method, a grating pattern (evenly spaced lines black and white lines) is attached to the specimen, and its image is captured using a digital camera before and after deformation. By digitally sampling the recorded image at an interval (thinning-out), a Moiré fringe pattern is generated [3]

2.3 Sampling Moire Method Mathematical Model

- If we start from 2nd pixel and did the thinning out, we will obtain same moire pattern with phase shift of $\frac{2\pi}{N}$
- Starting from kth pixel results in phase shift of $\frac{2\pi k}{N}$

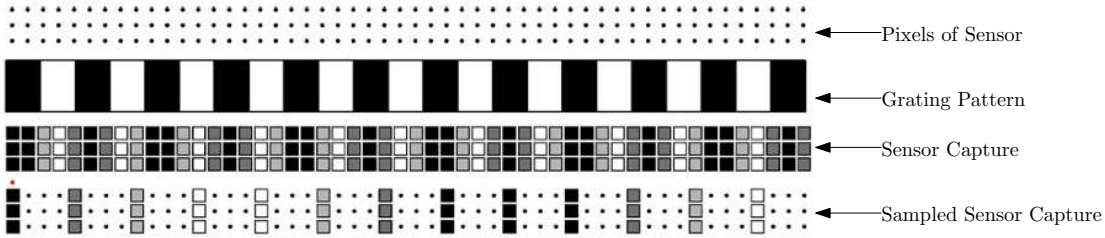


Figure 2.2: Thinning Out [3]

Figure 2.3: Resultant Pattern After Interpolation[3]

- Resultant Interpolated Moire pattern can be mathematically represented as

$$I_k(i) = I_a(i)\cos\left(\phi_m(i) + \frac{2\pi k}{N}\right) + I_b(i) \tag{2.1}$$

where I_b represents backlight, ϕ_m represents initial phase and it can be obtained by

$$\phi_m(i) = -\tan^{-1}\left(\frac{\sum_{k=0}^{N-1} I_k(i)\sin\left(\frac{k2\pi}{N}\right)}{\sum_{k=0}^{N-1} I_k(i)\cos\left(\frac{k2\pi}{N}\right)}\right) \tag{2.2}$$

2.4 Displacement Measurement

- The moire fringe phase ϕ_m calculated above can be considered as difference between specimen grating and reference grating $\phi_m = \phi_s - \phi_r$.
- When displacement of u = p (one pitch of grating) is done within the grating pattern, the thinned out pattern moves by U = Q. Therefore the amplification factor = $\frac{Q}{p}$,

$$U(fringe\ movement) = \frac{Q}{p} \cdot u(GratingMovement) \tag{2.3}$$

- Phase change (or displacement) of p in grating corresponds to phase change of 2π, which corresponds to phase change (or displacement) of Q in the Moire' Pattern.
- Displacement U in moire pattern corresponds to phase change of $\Delta\phi_m$ Therefore, $\frac{U}{Q} = \frac{\Delta\phi_m}{2\pi}$

$$u = p \cdot \frac{U}{Q} = p \cdot \frac{\Delta\phi_m}{2\pi} \tag{2.4}$$

- So measuring u boils down into measuring $\Delta\Phi_m$, which can be obtained by intensity values and Eq.2.1 above before and after deformation: $\Delta\phi_m = \phi'_m - \phi_m$.

Reference [3] Range of measurement is restricted to one pitch of the reference grating and the resolution of measurement is about $\frac{1}{1000}$ of the pitch (According to the ADC Resolution). If there's a provision to vary the pitch electronically, then we can obtain a large-range and high resolution displacement sensor, which is the key idea in development of the **Dynamic Moire Sensor**

2.5 Dynamic Moire Displacement Sensor

The Dynamic Moiré Displacement Sensor is based on the concept of dynamically varying the grating pitch before and after displacement measurement. A smaller grating pitch provides high-resolution displacement measurement, while a larger pitch allows measurement over a longer displacement range. By combining measurements obtained from gratings of different pitches, the system achieves both high resolution and long range measurement within a minimal optical setup. This approach enables accurate long-range displacement measurement while maintaining fine measurement sensitivity.

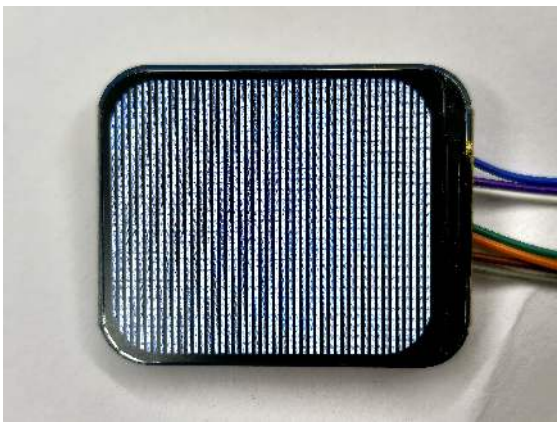


Figure 2.4: LCD Display

In this project, an LCD Display from Waveshare Electronics is used to generate dynamic reference grating and optical system is developed to detect the pattern.

Chapter 3

Sensor System Design

3.1 Dynamic Moiré Sensor Design

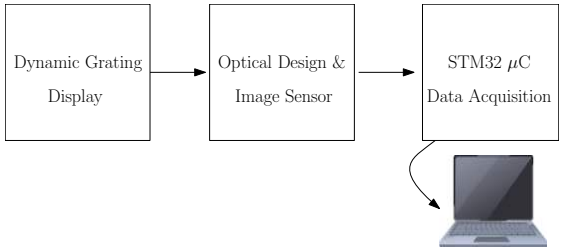


Figure 3.1: System Design Block Diagram

Dynamic Moire Sensor system design have 3 main parts: LCD Screen, Optical Setup and Sensing Hardware. The displayed dynamic pattern in LCD Screen is imaged by a convex lens to linear image sensor (S11108). With the custom driver circuit, CMOS Image sensor shifts the pixels data serially and sensed by STM32 micro-controller board. The data is then sent to Laptop via USB communication.

3.2 Optics Design and Sensor

Optics Design of Dynamic Moire Sensor is inspired by Line Scan Cameras where they use one convex lens to focus object to linear image sensors Fig. 3.2

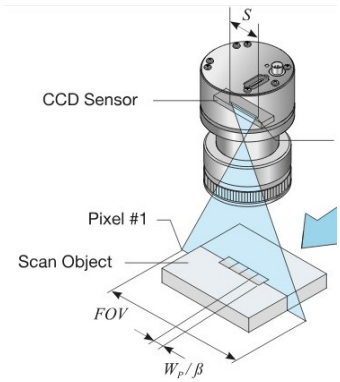


Figure 3.2: Line Scane Camera

Optics Simulation in OSLO

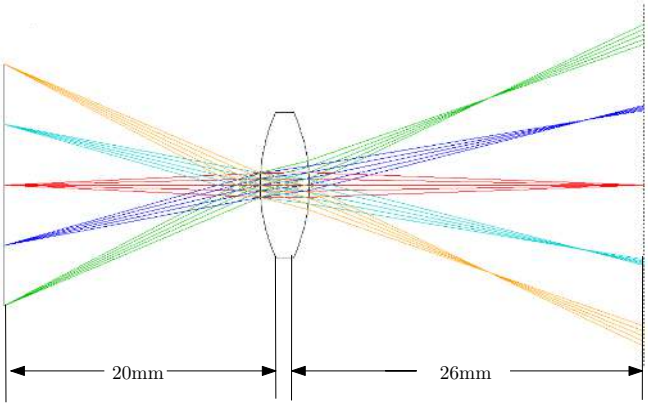


Figure 3.3: Optics Design in OSLO

Optical Design Simulation 3.3 is done in OSLO Software and we obtained the image distance and object distance from the convex lens and was implemented in our system. The OLED Display was kept below that to display dynamic grating pattern 3.4

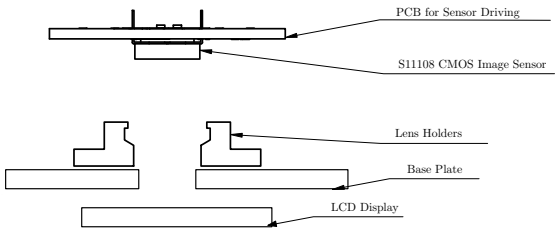
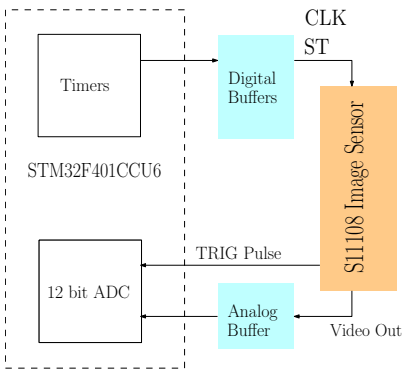


Figure 3.4: Hardware Design with optics design

In order to capture the image, a linear CMOS image sensor (S11108) from Hamamatsu is used Fig 3.5a and a driver circuit was developed to set the clock and integration time of the sensor. The timing requirements for the S11108 image

(a) S11108 Image Sensor



(b) S11108 Driver Circuit

Figure 3.5: S11108 linear image sensor and its driver circuit

sensor are illustrated in Fig. 3.6. A 1 MHz clock and a start pulse are applied to control the sensor operation and set the integration time. The integration time

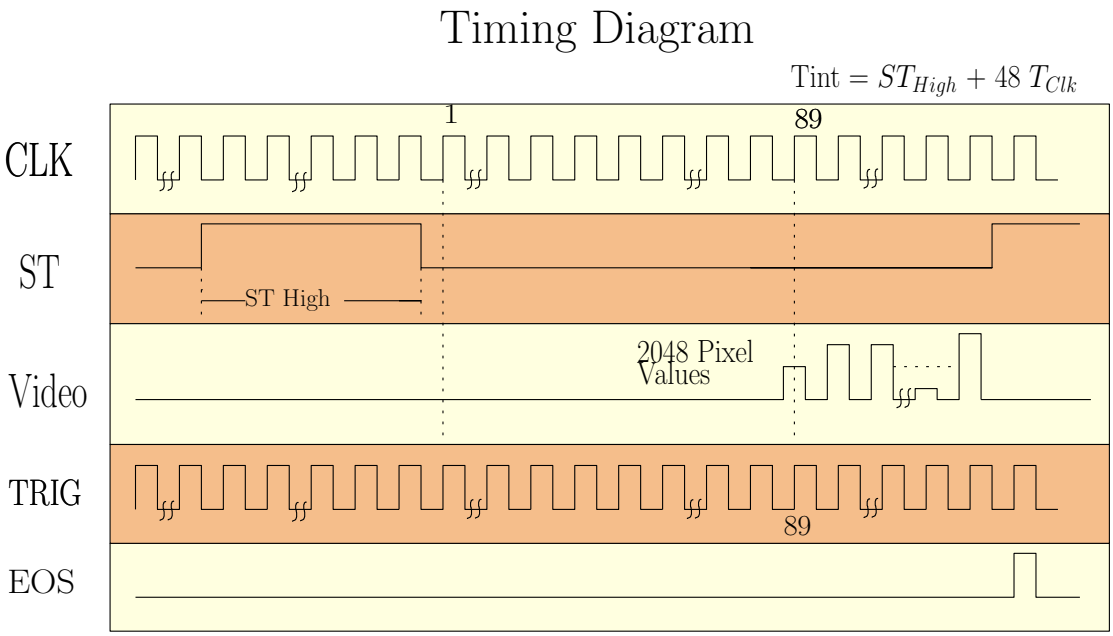


Figure 3.6: Timing Diagram for Sensor

is determined by the high duration of the start pulse. The sensor provides three output signals: Video, TRIG, and EOS (End of Scan). The analog pixel values from all 2048 pixels are sequentially shifted out through the video output and must be sampled at the positive edge of the TRIG signal. After all 2048 samples are read, the EOS signal generates a high pulse for one clock cycle, indicating the completion of the scan.

3.3 Hardware Design for Data Acquisition

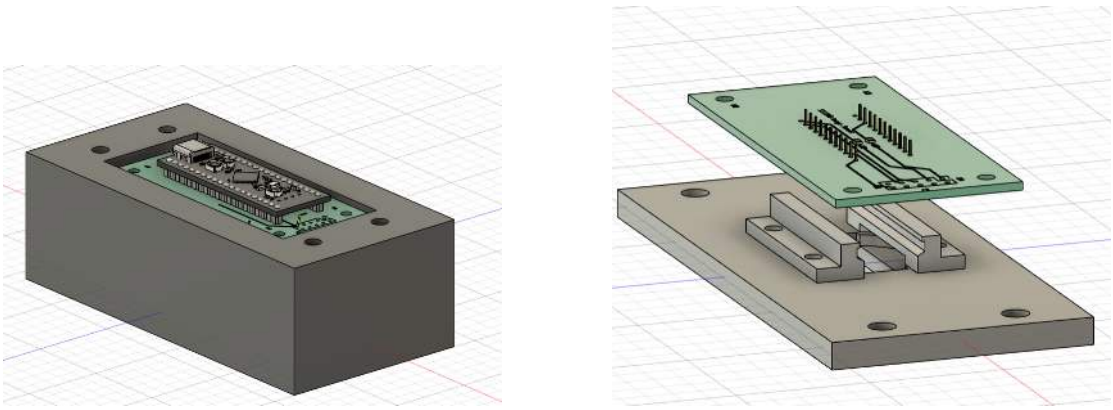


Figure 3.7: 3D Model Design

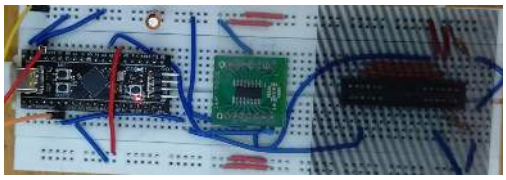
In the hardware design, PCBs containing the driver circuits and the optical arrangement were enclosed in a box to form the initial prototype of the sensor system as given in Fig 3.7

Chapter 4

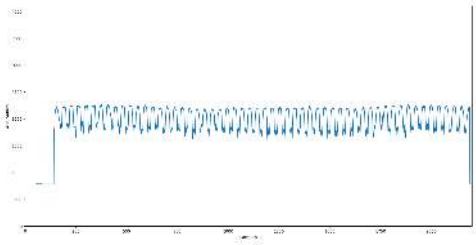
Experiments, Results & Analysis

4.1 Initial Results from Fixed Grating

Initial Results grating pattern capture from the CMOS Image Sensor was taken with breadboard setup in Fig. 4.1a and the grating pattern of 0.5 mm pitch is obtained as Fig. 4.1b

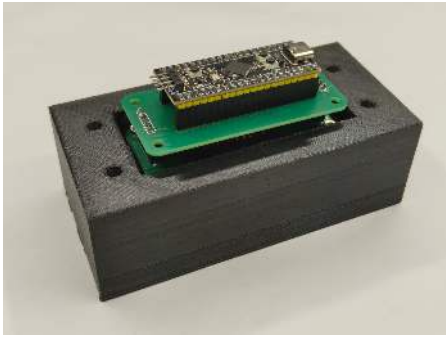


(a) Initial Breadboard Setup

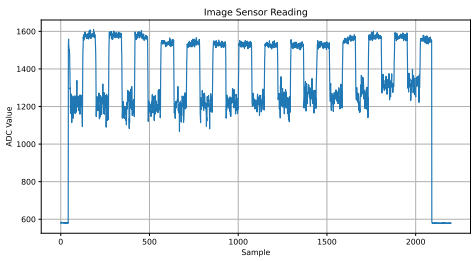


(b) Sensor Readings for 0.5mm pitch transparent grating

Figure 4.1: Experimental Setup for fixed grating



(a)



(b) Image Sensor Readings for 2mm pitch

Figure 4.2: Hardware Setup and Output for fixed grating

Next grating image was taken with 2mm pitch transparent grating and our new Dynamic Moire Hardware setup. We got resultant waveform as shown in 4.2b of the 2mm grating

4.2 Results from Dynamic Grating

Next Test was with LCD Display and Dynamic Moire Sensor Hardware as shown in Fig 4.3 The Raw ADC Pixels from the Sensor is shown in Fig 4.5 where we

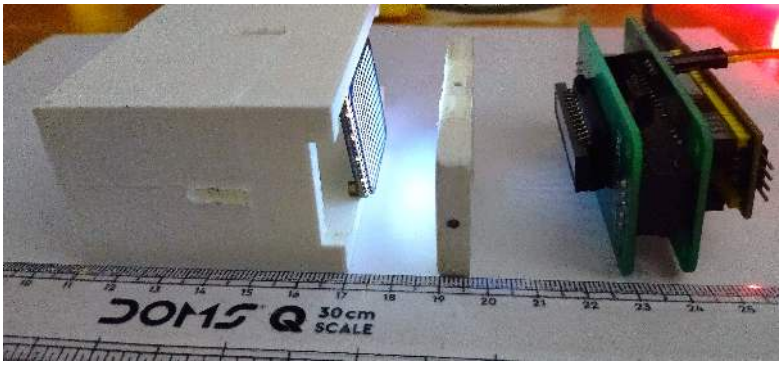


Figure 4.3: Dynamic Moire Sensor + LCD Display

can see distortion of the sinusoidal waveform representing black and white grating introduced due to vignetting effect of lens with smaller diameter.

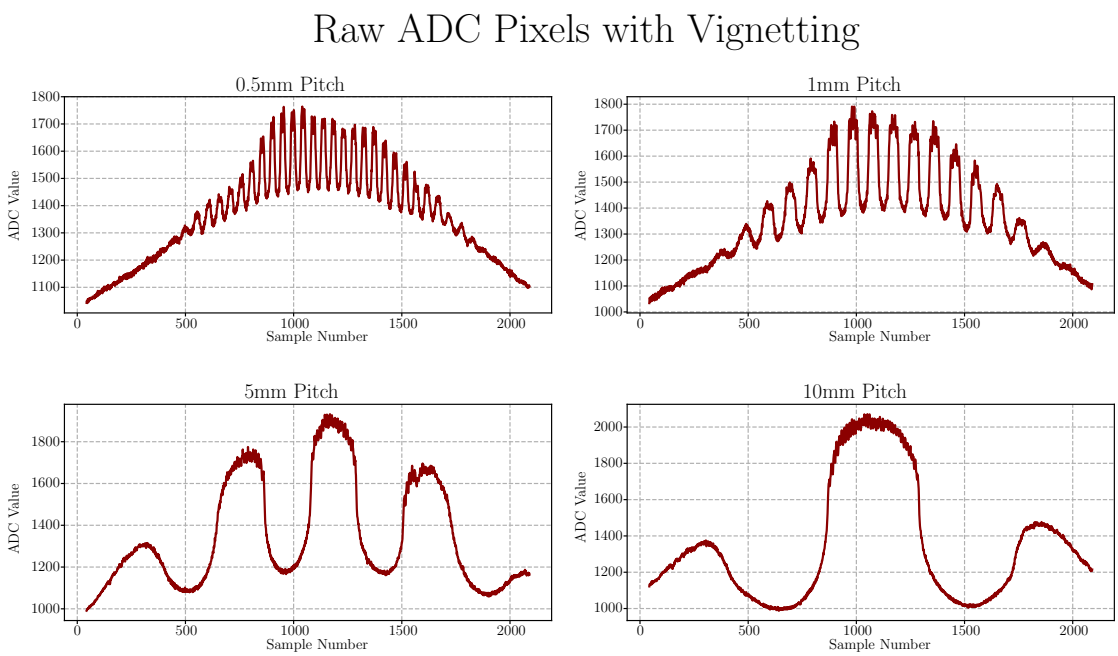


Figure 4.4: Grayscale Image of Signal

The non-uniform intensity observed across the sensor is mainly caused by vignetting in the optical system. To compensate for this effect, a flat-field correction is performed using the intensity profile obtained from a uniform white light image.

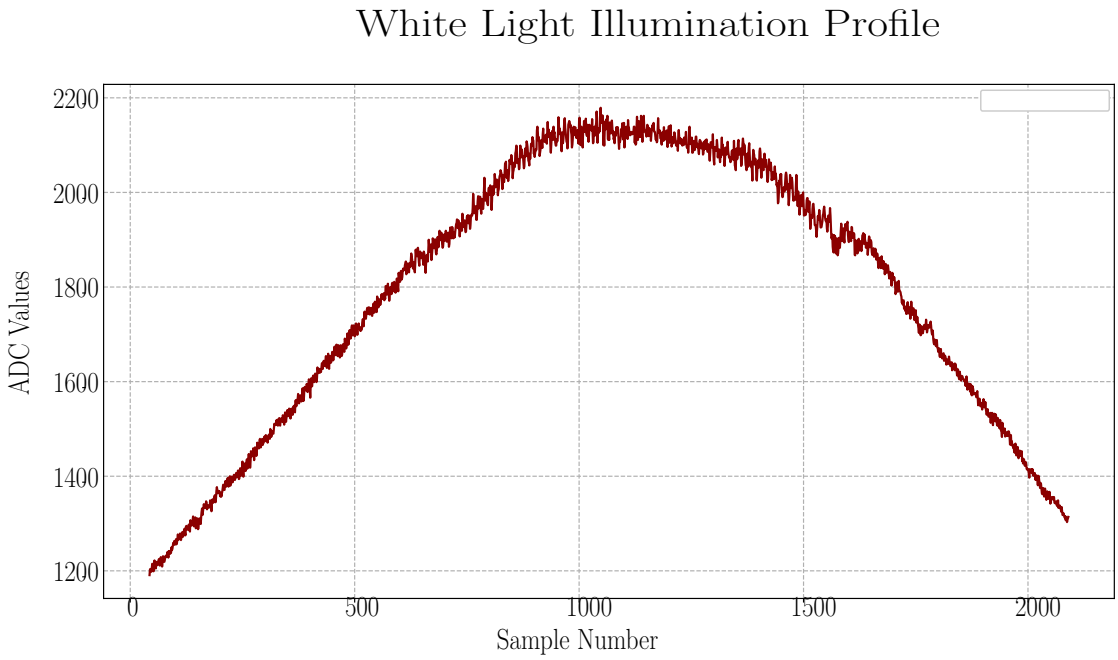


Figure 4.5: Grayscale Image White Light

To model this effect, a uniform white light image is captured by displaying a full white pattern on the LCD. Ideally, the captured intensity across the sensor should be constant. However, due to vignetting, the measured intensity profile shows a gradual reduction towards the edges. This measured white light profile is used as a reference to characterize the vignetting effect.

Let the measured signal from the sensor be represented as

$$S(x) = I(x) \cdot V(x)$$

$S(x)$ - measured sensor signal
$I(x)$ - actual optical signal of grating pattern
$V(x)$ - vignetting function

To correct the distortion introduced by vignetting, the measured signal is normalized using the obtained vignetting profile

$$I(x) = \frac{S(x)}{V(x)}$$

This correction compensates for the intensity falloff across the sensor and restores the true sinusoidal waveform of the Moiré signal. After vignetting correction, the signal amplitude becomes uniform on middle part, but towards the edges, the intensity could not be restored

Fig .4.6 and the corresponding grayscale image Fig .4.7

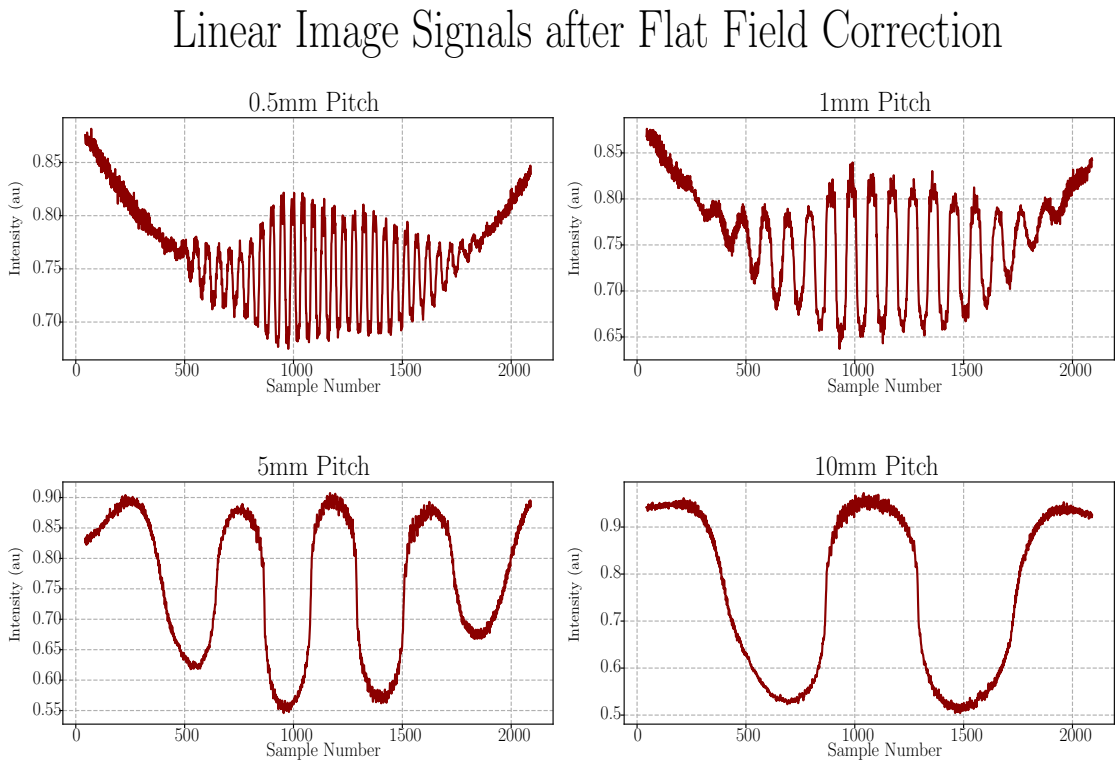


Figure 4.6: Signal After Flat Field Correction

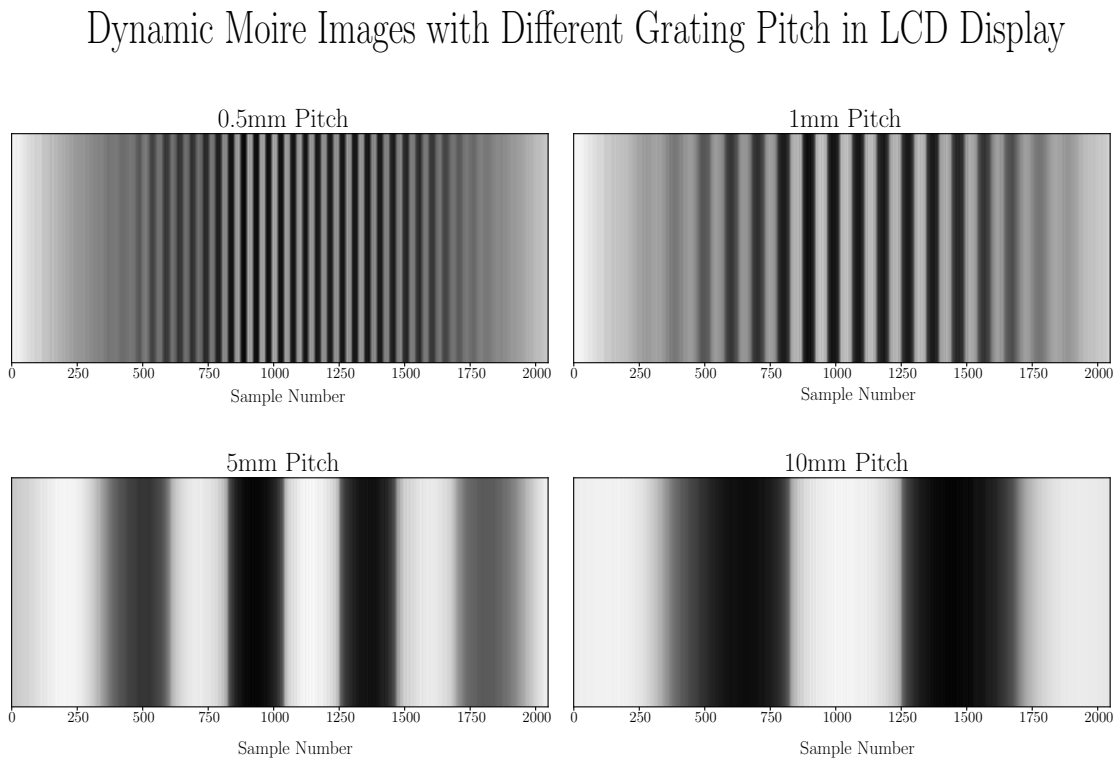


Figure 4.7: Grayscale Image of Signal

4.3 Conclusion and Future Perspective

The obtained images of grating patterns have aberration effects introduced due to convex lens. Displacement can still be calculated with the current data after

processing, which will be our future work for this project.
Another optical module which can be used to reduce aberrations are CIS Modules Fig 4.8, which are typically used in flatbed scanners. In the future prototypes this module can be included to obtain a clean pattern

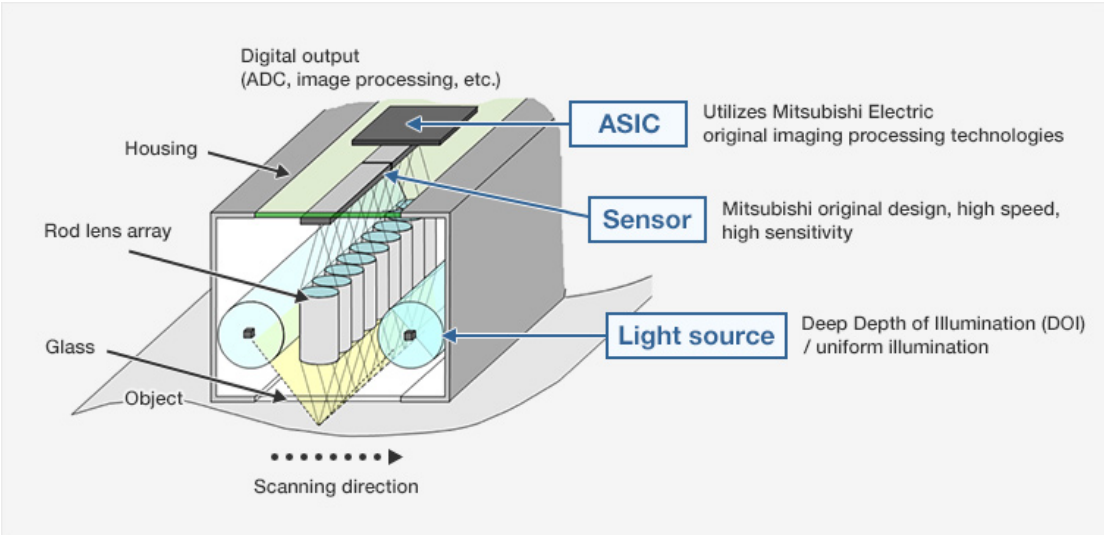


Figure 4.8: Contact Image Sensor [Mitsubishi CIS Modules]

Furthermore an X-Y Stage Displacement Measurement Setup with 6mm x 6mm image sensor can also be developed with the current lens system, as it will introduce lesser aberration effects in that distance

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